

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: _____

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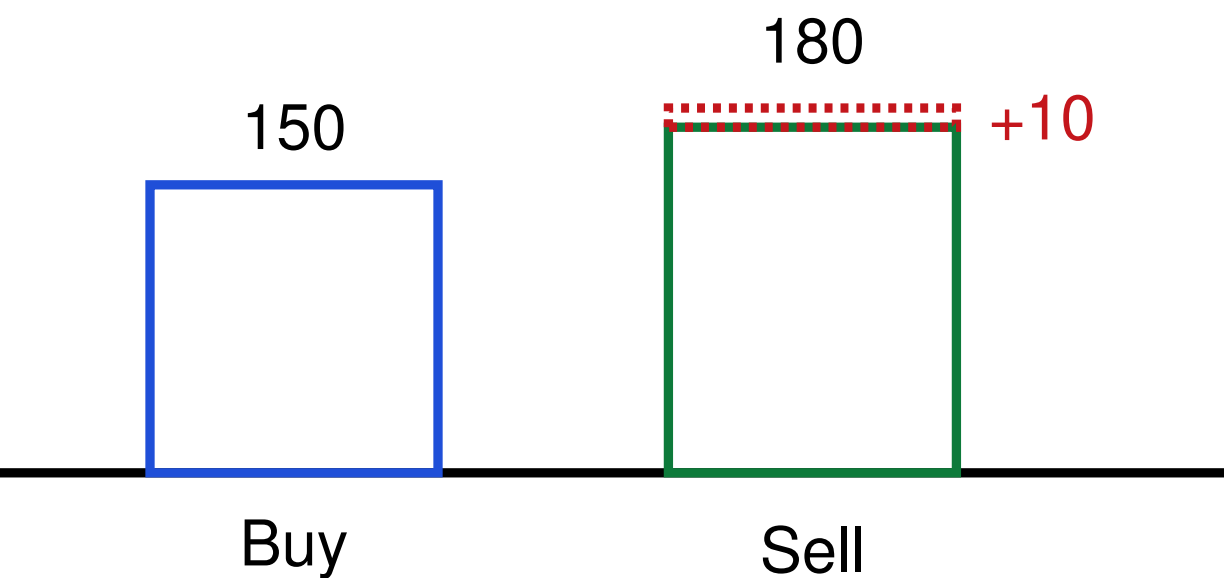
UNIT V: BASICS OF INVESTMENT

1. Investment Return

Gain or loss as a % of the money invested.

$$\begin{aligned}\text{Absolute return} &= (V_f - V_i) / V_i \times 100 \\ \text{HPR} &= [\text{Income} + (V_f - V_i)] / V_i \times 100 \\ \text{CAGR} &= (V_f / V_i)^{1/n} - 1\end{aligned}$$

Eg: Rs 150 → Rs 180, dividend Rs 10
HPR = (10 + 30) ÷ 150 = 26.67%



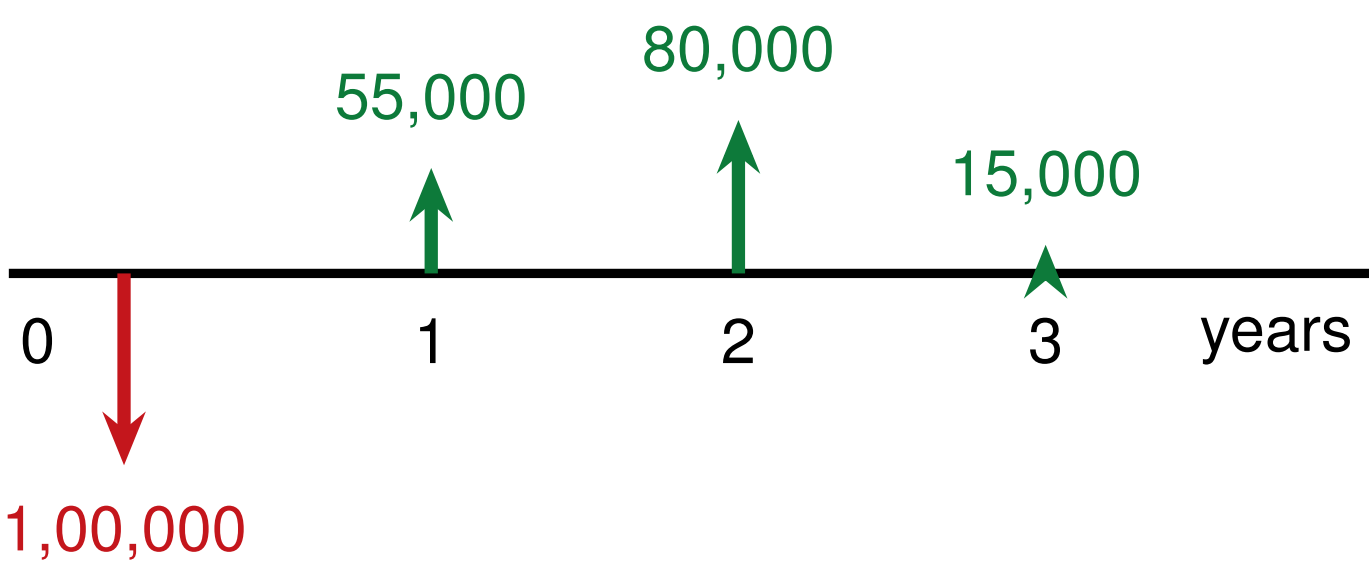
2. Uneven Cash Flows

Unequal cash flows: discount each to today.

$$\text{NPV} = \sum \frac{C_t}{(1+k)^t} - I$$

NPV > 0: accept; NPV < 0: reject.
IRR = rate at which NPV = 0.

Eg: I = Rs 1,00,000, k = 10%
NPV ≈ Rs 27,340 > 0, so accept



3. Compounding Frequency

More frequent interest = a little more money.

$$\begin{aligned}A &= P \left(1 + \frac{r}{n}\right)^{nt} & \text{Continuous: } A &= Pe^{rt} \\ \text{Effective annual rate} &= \left(1 + \frac{r}{n}\right)^n - 1\end{aligned}$$

Eg: Rs 5,00,000 at 8% for 1 year

Compounding	Amount (Rs)	EAR
Yearly	5,40,000	8.00%
Quarterly	5,41,216	8.24%
Monthly	5,41,500	8.30%
Daily	5,41,639	8.33%
Continuous	5,41,644	8.33%

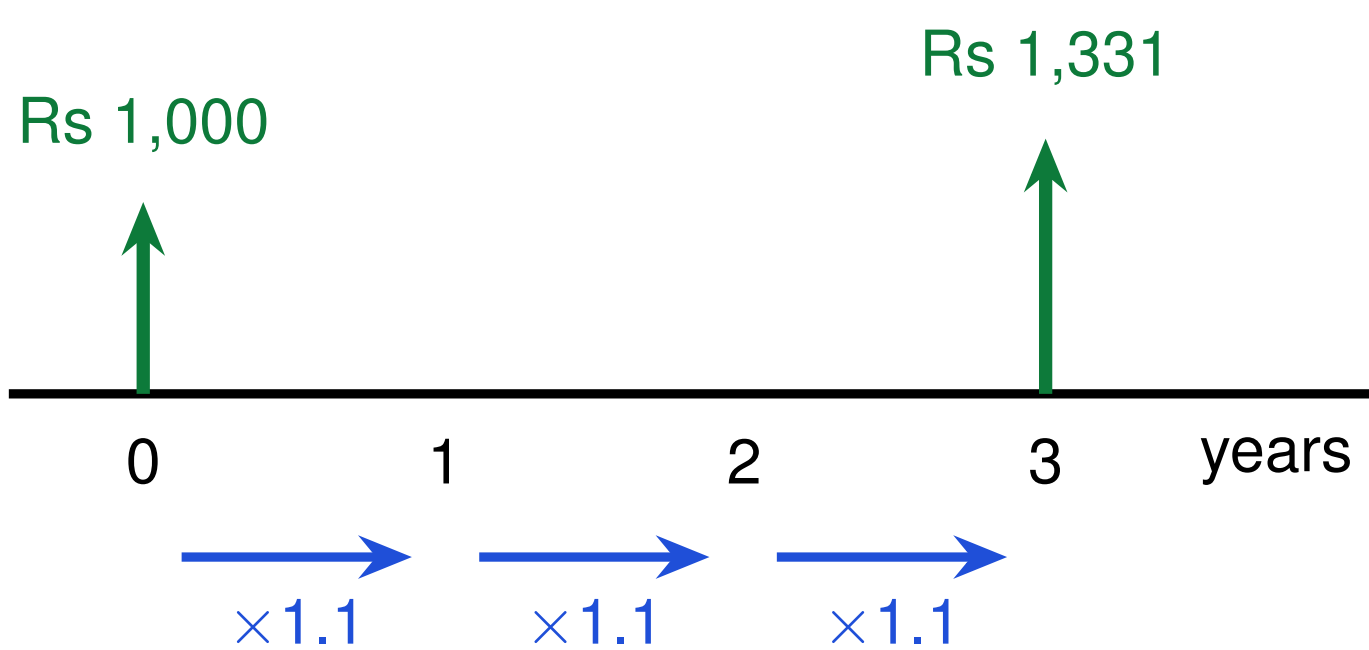
4. Economic Equivalence

Same value at a common point in time.

$$\begin{aligned}FV &= PV(1+i)^n \\ PV &= \frac{FV}{(1+i)^n}\end{aligned}$$

Eg: at 10%, Rs 1,000 today ≡ Rs 1,331 after 3 years

Use: compare loans, lease vs purchase.



UNIT VI: RISK AND UNCERTAINTY

1. Decision under Risk and Uncertainty

Risk = chance of an undesirable event that causes a financial loss.

Risk management = identify, assess and control risk.

Types of risk:

Strategic	Compliance
Operational	Financial

$$\begin{aligned}\text{Expected return } \bar{X} &= \sum X_i P(X_i) \\ \text{Variance } \sigma^2 &= \sum d^2 / n \quad (\text{or } \sum d^2 P) \\ \text{SD } \sigma &= \sqrt{\sigma^2}, \quad d = X_i - \bar{X} \\ \text{CV} &= \frac{\sigma}{\bar{X}} \times 100\end{aligned}$$

Eg: returns 12, 18, 20, 15, 25 (%)
mean 18%, σ = 4.43%, CV = 24.6%

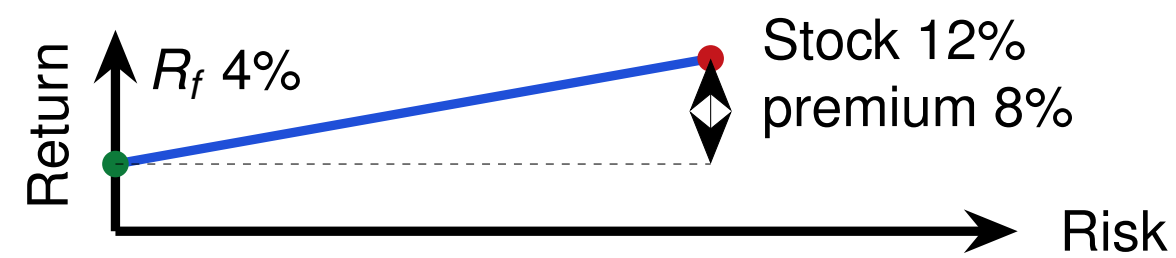
Higher σ or CV = riskier.

2. Risk Premium

Extra return for taking risk (vs risk-free).

$$\begin{aligned}\text{Risk premium} &= E(R) - R_f \\ \text{Sharpe ratio} &= \frac{R_p - R_f}{\sigma_p}\end{aligned}$$

Eg: E(R) = 12%, R_f = 4%, σ = 10%
Premium = 8%, Sharpe = 0.8

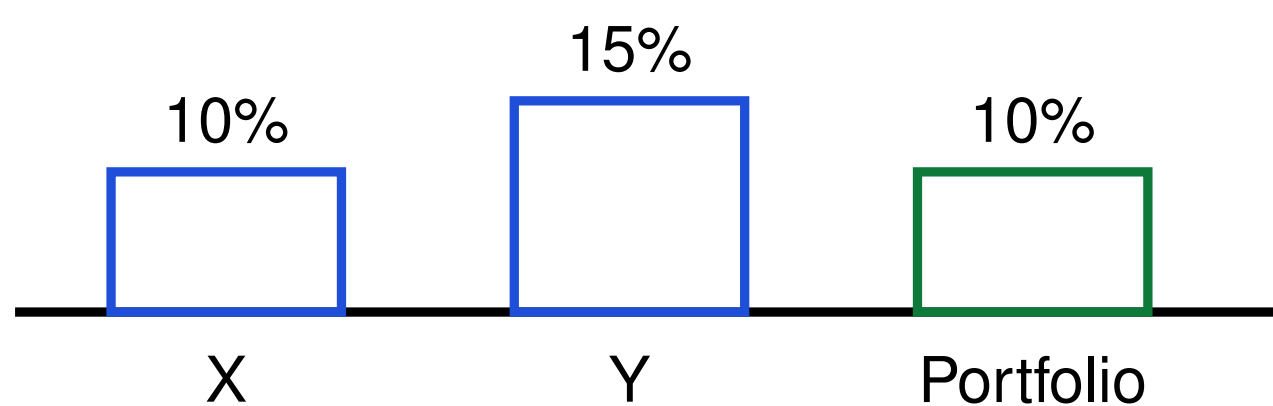


3. Portfolio Diversification

Spread over assets with ρ < 1 to lower risk.

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}$$

Eg: w = 0.5, 0.5; σ = 10%, 15%; ρ = 0.25
σ_p² = 0.01, so σ_p = 10%



4. Life Insurance

You pay a premium. The insurer pays the sum assured S if you die within the term.

$$\text{APV} = S \sum v^k P(T = k), \quad v = \frac{1}{1+i}$$

Eg: 3-yr term, S = Rs 2,00,000, i = 5%
P(T = 1, 2, 3) = 0.003, 0.0025, 0.002
APV = Rs 1,370.48

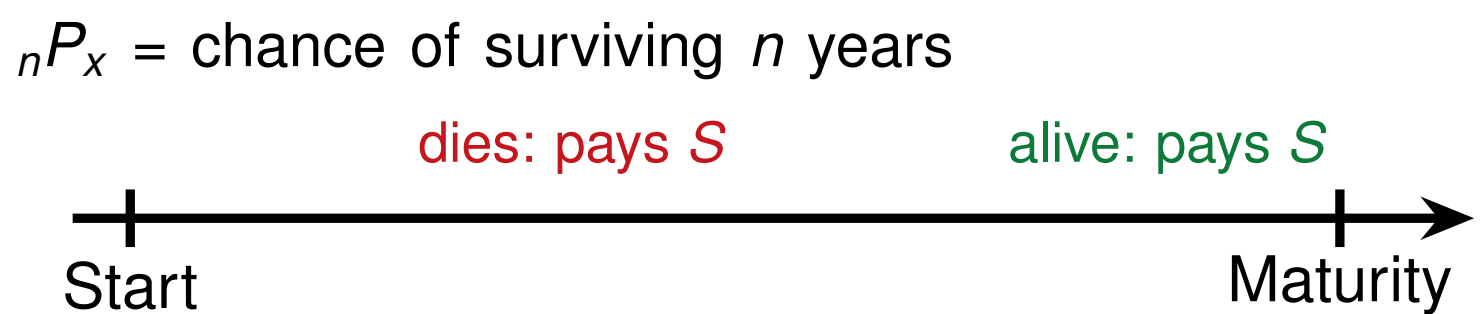


5. Endowment

Pays the sum assured S if you die in the term OR survive to maturity (protection + savings).

$$\text{APV} = S [\sum v^k P(T = k) + v^n {}_n P_x]$$

Eg: 20-yr plan, S = Rs 1,00,000;
Rs 1,00,000 paid on death or on survival



Investors demand a higher expected return for taking higher risk